

Test Plan

Automated Testing Framework for a Web Application

DEPI Graduation Project

1. Project Overview

This project focuses on designing and implementing an automated testing framework for a web application. It covers manual, automation, API, database, and performance testing using industry-standard tools within an Agile Scrum approach.

2. Tested Websites

Manual and Automation testing use the same website. Other testing types each use a dedicated system.

Testing Type	Website	URL
Manual & Automation	SauceDemo	https://www.saucedemo.com
API Testing	ReqRes	https://reqres.in
Database Testing	OpenCart Demo	https://demo.opencart.com
Performance Testing	SauceDemo	https://www.saucedemo.com

3. Tools Used

- **UI Automation:** Selenium + Page Object Model (POM)
- **API Testing:** Postman & Newman
- **Database Testing:** MySQL
- **Performance Testing:** JMeter
- **CI/CD:** Jenkins
- **BDD:** Cucumber (Gherkin feature files)
- **Test Management:** Jira
- **Version Control:** Git / GitHub

4. Entry Criteria

- Test environment is ready and accessible

- Target websites are stable and reachable
- Test data is prepared
- All tools are installed and configured

5. Exit Criteria

- All planned test cases have been executed
- Critical defects are fixed and re-tested
- Test results are documented
- Final report is completed

6. Sprint Schedule

Sprint 1 — Planning

- Define project scope and objectives
- Prepare Test Plan document
- Define entry and exit criteria
- Plan testing schedule and resources

Deliverables: *Test Plan, Project Roadmap*

Sprint 2 — Analysis & Design

- Perform exploratory testing
- Design test cases for UI, API, and database
- Identify automation candidates
- Create BDD feature files (Cucumber)
- Log test cases and defects in Jira

Deliverables: *Test Cases (Manual + Automated), Feature Files, Jira Setup*

Sprint 3 — Implementation

- Develop Selenium automation scripts using POM
- Execute API tests in Postman
- Write SQL queries for database validation
- Set up GitHub repository and version control
- Configure JMeter for performance testing

Deliverables: *Automation Scripts, Postman Collections, SQL Queries, GitHub Repo*

Sprint 4 — Execution & Completion

- Execute full test suite (UI, API, DB, Performance)
- Analyze results and generate reports
- Integrate CI/CD pipeline using Jenkins
- Verify exit criteria and finalize documentation

Deliverables: *Execution Report, Performance Report, Final Documentation, CI/CD Pipeline*

7. Test Deliverables

- Test Plan Document
- Test Cases (Manual & Automated)
- Bug Reports
- Automation Scripts
- API Collections
- Performance Reports
- Final Test Report

8. Agile Practices

- Daily Standups (15 min): what was done, what's planned, any blockers
- Sprint Planning: define goals and tasks at the start of each sprint
- Sprint Retrospective: review what went well, what didn't, and improvements